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Properties of Matter

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MCQ (Single Correct Answer)	82 questions
Answer key	at the end

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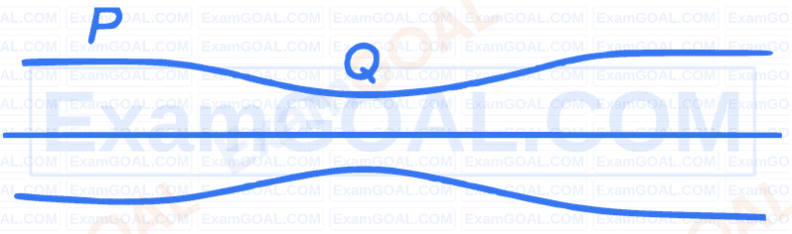
MCQ (Single Correct Answer) 82 questions

Q1. RE-NEET 2026

Open solution

Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is $15\mathrm{Nm}^{-2}$. The area of cross-section at P and Q are $40\ \mathrm{cm}^2$ and $20\ \mathrm{cm}^2$, respectively. The rate of flow of water through the pipe, in $\mathrm{cm}^3\ \mathrm{s}^{-1}$, is:

[Take density of water $= 1000\ \mathrm{kg\ m}^{-3}$]



(A) 400 (B) 100 (C) 200 (D) 300

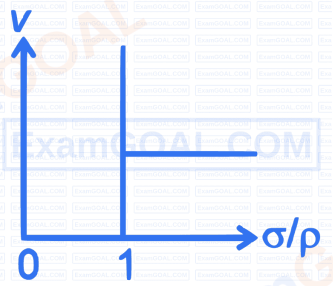
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Q2. RE-NEET 2026

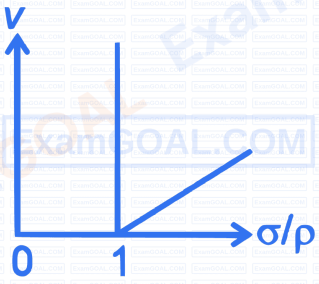
Open solution

In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by:

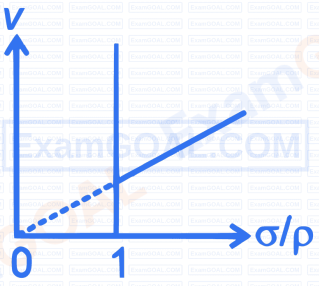
(A)



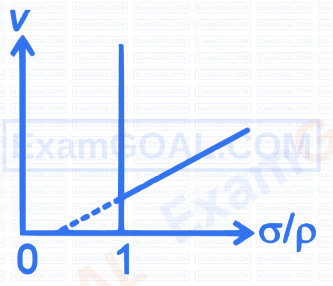
(B)



(C)



(D)



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Q3. NEET 2026

Open solution

Match List I with List II:

List I		List I I	
A.	Young's Modulus	I.	$\frac{\Delta d}{\Delta L}\left(\frac{L}{d}\right)$
B.	Compressibility	II.	$\frac{FL}{A(\Delta L)}$
C.	Bulk Modulus	III.	$-\frac{1}{\Delta P}\left(\frac{\Delta V}{V}\right)$
D.	Poisson's Ratio	IV.	$-P\left(\frac{V}{\Delta V}\right)$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-III, B-II, C-I, D-IV

(C) A-I, B-IV, C-III, D-II

(D) A-II, B-III, C-IV, D-I

Q4. NEET 2026

Open solution

A submarine is designed to withstand an absolute pressure of 100 atm . How deep can it go below the water surface?

(Consider the density of water $= 1000\ \mathrm{kg\ m^{-3}}$, $1\ \mathrm{atm} = 1 \times 10^5\ \mathrm{Pa}$ and gravitational acceleration $g = 10\ \mathrm{m/s^2}$)

(A) 990 m

(B) 9900 m

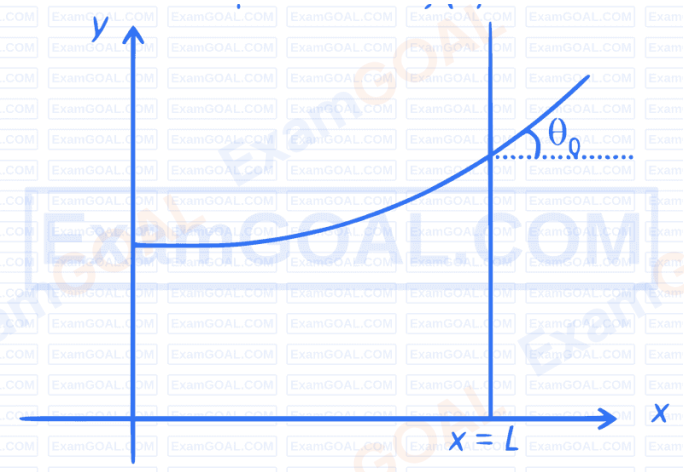
(C) 99 m

(D) 9000 m

Q5. NEET 2025

Open solution

Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ, the liquid surface makes angle θ_0 $(\theta_0 \ll 1)$ with the x-axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is:



(take $\theta(x) = \sin\theta(x) = \tan\theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)

(A) $\frac{d^2y}{dx^2} = \sqrt{\frac{\rho g}{S}}$

(B) $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}}x$

(C) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}x$

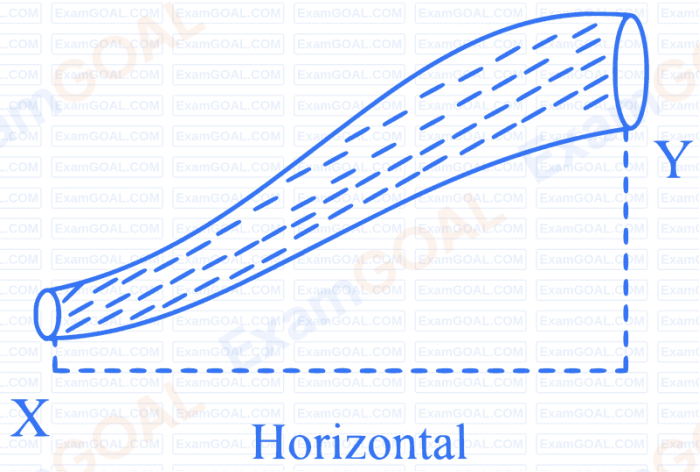
(D) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}y$

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Q6. NEET 2024 (Re-Examination)

An ideal fluid is flowing in a non-uniform cross-sectional tube XY (as shown in the figure) from end X to end Y. If K_1 and K_2 are the kinetic energy per unit volume of the fluid at X and Y respectively, then the correct option is :

Open solution



(A) $K_1 = K_2$ (B) $2K_1 = K_2$ (C) $K_1 > K_2$ (D) $K_1 < K_2$

Q7. NEET 2024

The maximum elongation of a steel wire of $1\ \mathrm{m}$ length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8\ \mathrm{N\,m^{-2}}$ and $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$, is:

(A) 4 mm (B) 0.4 mm (C) 40 mm (D) 8 mm

Open solution

Q8. NEET 2024

A thin flat circular disc of radius $4.5\ \mathrm{cm}$ is placed gently over the surface of water. If surface tension of water is $0.07\ \mathrm{N\,m^{-1}}$, then the excess force required to take it away from the surface is

(A) 19.8 mN (B) 198 N (C) 1.98 mN (D) 99 N

Open solution

Q9. NEET 2024

A metallic bar of Young's modulus, $0.5 \times 10^{11}\ \mathrm{N\,m^{-2}}$ and coefficient of linear thermal expansion $10^{-5}{}^{\circ}\mathrm{C}^{-1}$, length $1\ \mathrm{m}$ and area of cross-section $10^{-3}\ \mathrm{m}^2$ is heated from $0^{\circ}\mathrm{C}$ to $100^{\circ}\mathrm{C}$ without expansion or bending. The compressive force developed in it is :

(A) $5 \times 10^3\ \mathrm{N}$ (B) $50 \times 10^3\ \mathrm{N}$ (C) $100 \times 10^3\ \mathrm{N}$ (D) $2 \times 10^3\ \mathrm{N}$

Open solution

Q10. NEET 2023 Manipur

The amount of elastic potential energy per unit volume (in SI unit) of a steel wire of length $100\ \mathrm{cm}$ to stretch it by $1\ \mathrm{mm}$ is (if Young's modulus of the wire $= 2.0 \times 10^{11}\mathrm{Nm}^{-2}$) :

(A) 10^{11} (B) 10^{17} (C) 10^7 (D) 10^5

Open solution

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Q11. NEET 2023 Manipur

Which of the following statements is not true?

(A) Coefficient of viscosity is a scalar quantity

(B) Surface tension is a scalar quantity

(C) Pressure is a vector quantity

(D) Relative density is a scalar quantity

Open solution

Q12. NEET 2023 Manipur

The viscous drag acting on a metal sphere of diameter $1\ \text{mm}$, falling through a fluid of viscosity $0.8\ \text{Pa s}$ with a velocity of $2\ \text{m s}^{-1}$ is equal to :

(A) $15 \times 10^{-3}\ \text{N}$ (B) $30 \times 10^{-3}\ \text{N}$ (C) $1.5 \times 10^{-3}\ \text{N}$ (D) $20 \times 10^{-3}\ \text{N}$

Open solution

Q13. NEET 2023

The amount of energy required to form a soap bubble of radius $2\ \text{cm}$ from a soap solution is nearly: (surface tension of soap solution $= 0.03\ \text{N m}^{-1}$)

(A) $5.06 \times 10^{-4}\ \text{J}$ (B) $3.01 \times 10^{-4}\ \text{J}$ (C) $50.1 \times 10^{-4}\ \text{J}$ (D) $30.16 \times 10^{-4}\ \text{J}$

Open solution

Q14. NEET 2023

The venturi-meter works on :

(A) Bernoulli's principle

(B) The principle of parallel axes

(C) The principle of perpendicular axes

(D) Huygen's principle

Open solution

Q15. NEET 2023

Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :

(A) W/A (B) $W/2A$ (C) Zero (D) $2W/A$

Open solution

Q16. NEET 2022 Phase 2

Two copper vessels A and B have the same base area but of different shapes. A takes twice the volume of water as that B requires to fill upto a particular common height. Then the correct statement among the following is :

(A) Vessel B weighs twice that of A.

(B) Pressure on the base area of vessels A and B is same.

(C) Pressure on the base area of vessels A and B is not same.

(D) Both vessels A and B weigh the same.

Open solution

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Q17. NEET 2022 Phase 2

The terminal velocity of a copper ball of radius 5 mm falling through a tank of oil at room temperature is 10 cm s^{-1}. If the viscosity of oil at room temperature is 0.9 kg m^{-1} s^{-1}, the viscous drag force is :

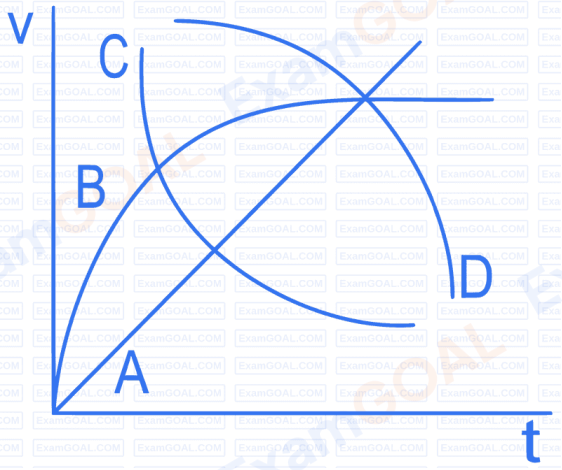
Open solution

(A) 4.23×10^{-6} N (B) 8.48×10^{-3} N (C) 8.48×10^{-5} N (D) 4.23×10^{-3} N

Q18. NEET 2022 Phase 1

A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is

Open solution



(A) A (B) B (C) C (D) D

Q19. NEET 2022 Phase 1

If a soap bubble expands, the pressure inside the bubble

Open solution

(A) Decreases (B) Increases
(C) Remains the same (D) Is equal to the atmospheric pressure

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Q20. NEET 2022 Phase 1

Open solution

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The stretching of a spring is determined by the shear modulus of the material of the spring.

Reason (R) : A coil spring of copper has more tensile strength than a steel spring of same dimensions.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q21. NEET 2021

Open solution

The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerin becomes constant after some time. If the density of glycerin is $\frac{d}{2}$, then the viscous force acting on the ball will be :

(A) 2Mg (B) $\frac{Mg}{2}$ (C) Mg (D) $\frac{3}{2}$Mg

Q22. NEET 2020 Phase 1

Open solution

A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5 g. Another capillary tube of radius 2r is immersed in water. The mass of water that will rise in this tube is :

(A) 5.0 g (B) 10.0 g (C) 20.0 g (D) 2.5 g

Q23. NEET 2020 Phase 1

Open solution

A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changed to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :

(A) $\frac{Mg(L_1-L)}{AL}$ (B) $\frac{MgL}{AL_1}$

(C) $\frac{MgL}{A(L_1-L)}$ (D) $\frac{MgL_1}{AL}$

Q24. NEET 2019

Open solution

When a block of mass M is suspended by a long wire of length L, the length of the wire becomes (L + l). The elastic potential energy stored in the extended wire is :

(A) Mgl (B) mgL (C) $\frac{1}{2}$Mgl (D) $\frac{1}{2}$MgL

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Q25. NEET 2019

A small hole of area of cross-section 2 mm^2 present near the bottom of a fully filled open tank of height 2 m. Taking g = 10 m/s^2, the rate of flow of water through the open hole would be nearly :

(A) $8.9 \times 10^{-6}\ m^3/s$ (B) $2.23 \times 10^{-6}\ m^3/s$ (C) $6.4 \times 10^{-6}\ m^3/s$ (D) $12.6 \times 10^{-6}\ m^3/s$

Open solution

Q26. NEET 2019

The unit of thermal conductivity is :

(A) $J\ m^{-1}\ K^{-1}$ (B) $W\ m\ K^{-1}$ (C) $J\ m\ K^{-1}$ (D) $W\ m^{-1}\ K^{-1}$

Open solution

Q27. NEET 2019

A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals the pressure at a point Z_0 below the free surface of water in a container. Taking g = 10 m/s^2, density of water = 10^3 kg/m^3, the value of Z_0 is :

(A) 1 cm (B) 10 cm (C) 100 cm (D) 0.5 cm

Open solution

Q28. NEET 2018

A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to

(A) r^3 (B) r^2 (C) r^5 (D) r^4

Open solution

Q29. NEET 2018

Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

(A) 6F (B) F (C) 9F (D) 4F

Open solution

Q30. NEET 2017

A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be

(A) 450 (B) 1000 (C) 1800 (D) 225

Open solution

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Q31. NEET 2017

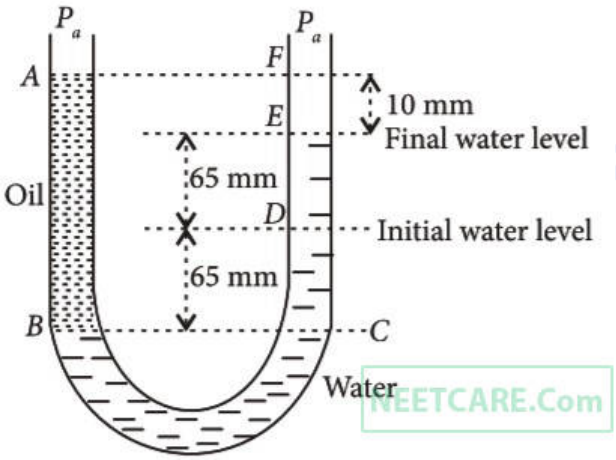
The bulk modulus of a spherical object is 'B'. If it is subjected to uniform pressure 'P', the fractional decrease in radius is

(A) $\frac{B}{3p}$ (B) $\frac{3p}{B}$ (C) $\frac{p}{3B}$ (D) $\frac{p}{B}$

Open solution

Q32. NEET 2017

A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is



(A) 425 kg m^{-3} (B) 800 kg m^{-3} (C) 928 kg m^{-3} (D) 650 kg m^{-3}

Open solution

Q33. NEET 2016 Phase 2

A rectangular film of liquid is extended from (4 cm $\times$ 2 cm) to (5 cm $\times$ 4 cm). If the work done is 3 $\times$ 10^{-4} J, the value of the surface tension of the liquid is

(A) 0.250 N m^{-1} (B) 0.125 N m^{-1} (C) 0.2 N m^{-1} (D) 8.0 N m^{-1}

Open solution

Q34. NEET 2016 Phase 2

Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T, rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey

(A) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$ (B) $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

(C) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$ (D) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

Open solution

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Q35. NEET 2016 Phase 2

Two identical bodies are made of a material for which the heat capacity increases with temperature. One of these is at 100°C, while the other one is at 0°C. If the two bodies are brought into contact, then, assuming no heat loss, the final common temperature is

Open solution

(A) 50 °C
(B) more than 50 °C
(C) less than 50 °C but greater than 0 °C
(D) 0 °C

Q36. NEET 2016 Phase 2

A body cools from a temperature 3T to 2T in 10 minutes. The room temperature is T. Assume that Newton's law of cooling is applicable. The temperature of the body at the end of next 10 minutes will be

Open solution

(A) $\frac{7}{4}$ T
(B) $\frac{3}{2}$ T
(C) $\frac{4}{3}$ T
(D) T

Q37. NEET 2016 Phase 1

Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_1 - l_2)$ is maintained same at all temperatures, which one of the following relations holds good?

Open solution

(A) $\alpha_1 l_2 = \alpha_2 {}^2 l_1$
(B) $\alpha_1 l_1 = \alpha_2 l_2$
(C) $\alpha_1 l_2 = \alpha_2 l_1$
(D) $\alpha_1 l_2^2 = \alpha_2 l_1^2$

Q38. NEET 2016 Phase 1

A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h [Latent heat of ice is 3.4×10^5 J/kg and g = 10 N/Kg]

Open solution

(A) 136 km
(B) 68 km
(C) 34 km
(D) 544 km

Q39. NEET 2016 Phase 1

A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1, at wavelength 500 nm is U_2 and that at 1000 nm is U_3. Wien's constant, b = 2.88×10^6 nm K. Which of the following is correct ?

Open solution

(A) $U_1 > U_2$
(B) $U_2 > U_1$
(C) $U_1 = 0$
(D) $U_3 = 0$

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Q40. NEET 2016 Phase 1

Two non-mixing liquids of densities ρ and nρ (n > 1) are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL (p < 1) in the denser liquid. The density d is equal to

Open solution

(A) $\{2+(n-1)p\}\rho$ (B) $\{1+(n-1)p\}\rho$

(C) $\{1+(n+1)p\}\rho$ (D) $\{2+(n+1)p\}\rho$

Q41. AIPMT 2015

The cylindrical tube of a spray pump has radius R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is

Open solution

(A) $\frac{VR^2}{n^3r^2}$ (B) $\frac{V^2R}{nr}$ (C) $\frac{VR^2}{n^2r^2}$ (D) $\frac{VR^2}{nr^2}$

Q42. AIPMT 2015

Water rises to a height h in capillary tube. If the length of capillary tube above the surface of water is made less than h, then

Open solution

(A) water rises upto a point a little below the top and stays there.

(B) water does not rise at all.

(C) water rises upto the tip of capillary tube and then starts overflowing like a fountain.

(D) water rises upto the top of capillary tube and stays there without overflowing.

Q43. AIPMT 2015

The value of coefficient of volume expansion of glycerin is $5 \times 10^{-4}\ K^{-1}$. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is

Open solution

(A) 0.025 (B) 0.010 (C) 0.015 (D) 0.020

Q44. AIPMT 2015

The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of

Open solution

(A) 4 : 1 (B) 1 : 1 (C) 1 : 2 (D) 2 : 1

Q45. AIPMT 2015 Cancelled Paper

The two ends of a metal rod are maintained at temperatures 100°C and 110°C. The rate of heat flow in the rod is found to be 4.0 J/s. If the ends are maintained at temperatures 200°C and 210°C, the rate of heat flow will be

Open solution

(A) 8.0 J/s (B) 4.0 J/s (C) 44.0 J/s (D) 16.8 J/s

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Q46. AIPMT 2015 Cancelled Paper

A wind with speed 40 m/s blows parallel to the roof of a house. The area of the roof is 250 m^2. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be $(\rho_{air} = 1.2 kg/m^3)$

Open solution

(A) 2.4×10^5 N, upwards
(B) 2.4×10^5 N, downwards
(C) 4.8×10^5 N, downwards
(D) 4.8×10^5 N, upwards

Q47. AIPMT 2015 Cancelled Paper

On observing light from three different starts P, Q and R, it was found that intensity of violet colour is maximum in the spectrum of P, the intensity of green colour is maximum in the spectrum of R and the intensity of red colour is maximum in the spectrum of Q. If T_P, T_Q and T_R are the respective absolute temperatures of P, Q and R, then it can be conclued from the above observations that

Open solution

(A) $T_P < T_R < T_Q$ (B) $T_P < T_Q < T_R$ (C) $T_P > T_Q > T_R$ (D) $T_P > T_R > T_Q$

Q48. AIPMT 2015 Cancelled Paper

The approximate depth of an ocean is 2700 m. The compressiblity of water is 45.4×10^{-11} Pa^{-1} and density of water is 10^3 kg/m^3. What fractional compression of water will be obtained at the bottom of the ocean?

Open solution

(A) 1.2×10^{-2} (B) 1.4×10^{-2} (C) 0.8×10^{-2} (D) 1.0×10^{-2}

Q49. AIPMT 2014

Copper of fixed volume V is drawn into wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. Which of the following graphs is a straight line ?

Open solution

(A) Δl versus $1/l$ (B) Δl versus l^2 (C) Δl versus $1/l^2$ (D) Δl versus l

Q50. AIPMT 2014

A certain number of spherical drops of a liquid of radius r coalesce to form a single drop of radius R and volume V. If T is the surface tension of the liquid, then

Open solution

(A) energy = 4VT $\left(\frac{1}{r} - \frac{1}{R}\right)$ is released.
(B) energy = 3VT$\left(\frac{1}{r} + \frac{1}{R}\right)$ is absorbed.
(C) energy = 3VT$\left(\frac{1}{r} - \frac{1}{R}\right)$ is released
(D) energy is neither released nor absorbed.

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Q51. AIPMT 2014

Steam at 100°C is passed into 20 g of water at 10°C. When water acquires a temperature of 80°C, the mass of water present will be

[Take specific heat of water = 1 cal g^{-1} $^oC^{-1}$ and latent heat of steam = 540 cal g^{-1}]

(A) 24 g (B) 31.5 g (C) 42.5 g (D) 22.5 g

Open solution

Q52. AIPMT 2014

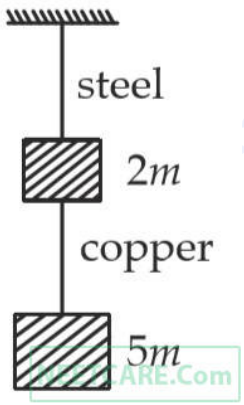
Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and to 54°C in the next 5 minutes. The temperature of the surroundings is

(A) 45°C (B) 20°C (C) 42°C (D) 10°C

Open solution

Q53. NEET 2013 (Karnataka)

If the ratio of diameters, lengths and Young's modulus of steel and copper wires shown in the figure are p, q and s respectively, then the corresponding ratio of increase in their lengths would be



(A) $\left(\frac{5q}{7sp^2}\right)$ (B) $\left(\frac{7q}{5sp^2}\right)$

(C) $\left(\frac{2q}{5sp}\right)$ (D) $\left(\frac{7q}{5sp}\right)$

Open solution

Q54. NEET 2013 (Karnataka)

Two metal rods 1 and 2 of same lengths have same temperature difference between their ends. Their thermal conductivities are K_1 and K_2 and cross sectional areas A_1 and A_2, respectively. If the rate of heat conduction in 1 is four times that in 2, then

(A) $K_1A_1 = 4K_2A_2$ (B) $K_1A_1 = 2K_2A_2$ (C) $4K_1A_1 = K_2A_2$ (D) $K_1A_1 = K_2A_2$

Open solution

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Q55. NEET 2013 (Karnataka)

A fluid is in streamline flow across a horizontal pipe of variable area of cross section. For this which of the following statements is correct?

(A) The velocity is maximum at the narrowest part of the pipe and pressure is maximum at the widest part of the pipe.

(B) Velocity and pressure both are maximum at the narrowest part of the pipe.

(C) velocity and pressure both are maximum at the widest part of the pipe.

(D) The velocity is minimum at the narrowest part of the pipe and the pressure is minimum at the widest part of the pipe.

Open solution

Q56. NEET 2013 (Karnataka)

The density of water at 20^oC is 998 kg/m^3 and at 40^oC is 992 kg/m^3. The coefficient of volume expansion of water is

(A) $3 \times 10^{-4}/^oC$ (B) $2 \times 10^{-4}/^oC$ (C) $6 \times 10^{-4}/^oC$ (D) $10 \times 10^{-4}/^oC$

Open solution

Q57. NEET 2013

A piece of iron is heated in a flame. It first becomes dull red then becomes reddish yellow and finally turns to white hot. The correct explanation for the above observation is possible by using

(A) Kirchhoff's Law

(B) Newton's Law of cooling

(C) Stefan's Law

(D) Wien's displacement Law

Open solution

Q58. NEET 2013

The wettability of a surface by a liquid depends primarily on

(A) density

(B) angle of contact between the surface and the liquid

(C) viscosity

(D) surface tension

Open solution

Q59. NEET 2013

The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?

(A) length = 200 cm, diameter = 2 mm

(B) length = 300 cm, diameter = 3 mm

(C) length = 50 cm, diameter = 0.5 mm

(D) length = 100 cm, diameter = 1 mm

Open solution

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Q60. NEET 2013

The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_p and C_v, respectively. If $\gamma = \frac{C_p}{C_v}$ and R is the universal gas constant, then C_v is equal to

Open solution

(A) $\frac{(\gamma-1)}{R}$
(B) γR
(C) $\frac{1+\gamma}{1-\gamma}$
(D) $\frac{R}{(\gamma-1)}$

Q61. AIPMT 2012 Mains

A slab of stone of area 0.36 m^2 and thickness 0.1 m is exposed on the lower surface to steam at 100°C. A block of ice at 0°C rests on the upper surface of the slab. In one hour 4.8 kg of ice is melted. The thermal conductivity of slab is
(Given latent heat of fusion of ice = 3.36×10^5 J kg^{-1})

Open solution

(A) 1.24 J/m/s/°C
(B) 1.29 J/m/s/°C
(C) 2.05 J/m/s/°C
(D) 1.02 J/m/s/°C

Q62. AIPMT 2012 Prelims

If the radius of a star is R and it acts as a black body, what would be the temperature of the star, in which the rate of energy production is Q?

Open solution

(A) $\frac{Q}{4\pi R^2\sigma}$
(B) $\left(\frac{Q}{4\pi R^2\sigma}\right)^{-1/2}$
(C) $\left(\frac{4\pi R^2 Q}{\sigma}\right)^{1/4}$
(D) $\left(\frac{Q}{4\pi R^2\sigma}\right)^{1/4}$

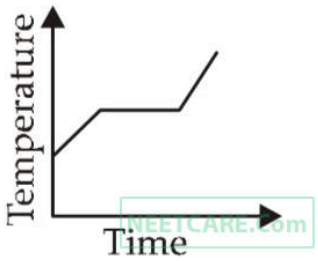
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Q63. AIPMT 2012 Prelims

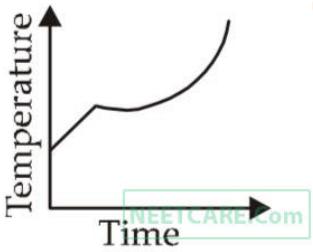
Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time ?

Open solution

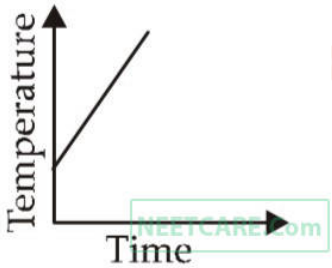
(A)



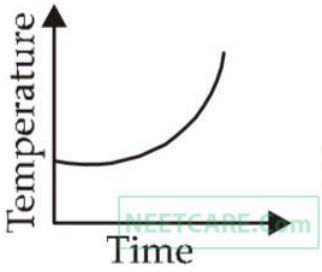
(B)



(C)



(D)



Q64. AIPMT 2010 Prelims

A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t. The metallic rod is melted and the material is formed into a rod of half the radius of the original rod. What is the amount of heat conducted by the new rod, when placed in thermal contact with the two reservoirs in time t?

Open solution

(A) $\frac{Q}{4}$ (B) $\frac{Q}{16}$ (C) 2Q (D) $\frac{Q}{2}$

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Q65. AIPMT 2010 Prelims

The total radiant energy per unit area, normal to the direction of incidence, received at a distance R from the centre of a star of radius r, whose outer surface radiates as a black body at a temperature TK is given by

Open solution

(A) $\frac{\sigma r^2 T^4}{R^2}$
(B) $\frac{\sigma r^2 T^4}{4\pi R^2}$
(C) $\frac{\sigma r^2 T^4}{R^4}$
(D) $\frac{4\pi\sigma r^2 T^4}{R^2}$

Q66. AIPMT 2010 Prelims

Assuming the sun to have a spherical outer surface of radius r, radiating like a black body at temperature t^{o}C, the power received by a unit surface, (normal to the incident rays) at a distance R from the centre of the sun is
where σ is the Stefan's constant.

Open solution

(A) $\frac{r^2\sigma(t+273)^4}{4\pi R^2}$
(B) $\frac{16\pi^2 r^2\sigma t^4}{R^2}$
(C) $\frac{r^2\sigma(t+273)^4}{R^2}$
(D) $\frac{4\pi r^2\sigma t^4}{R^2}$

Q67. AIPMT 2009

A black body at 227°C radiates heat at the rate of 7 cals/cm^{2}s. At a temperature of 727°C, the rate of heat radiated in the same units will be

Open solution

(A) 50
(B) 112
(C) 80
(D) 60

Q68. AIPMT 2009

The two ends of a rod of length L and a uniform cross-sectional area A are Kept at two temperatures T_1 and T_2 ($T_1 > T_2$). The rate of heat transfer, $\frac{dQ}{dt}$ through the rod in a steady state is given by :

Open solution

(A) $\frac{dQ}{dt} = \frac{k(T_1-T_2)}{LA}$
(B) $\frac{dQ}{dt} = kLA(T_1 - T_2)$
(C) $\frac{dQ}{dt} = \frac{kA(T_1-T_2)}{L}$
(D) $\frac{dQ}{dt} = \frac{kL(T_1-T_2)}{A}$

Q69. AIPMT 2008

On a new scale of temperature (which is linear) and called the W scale, the freezing and boiling points of water are 39°W and 239°W respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale ?

Open solution

(A) 200°W
(B) 139°W
(C) 78°W
(D) 117°W

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Q70. AIPMT 2007

Assuming the sun to have a spherical outer surface of radius r, radiating like a black body at temperature t^{o}C, the power received by a unit surface, (normal to the incident rays) at a distance R from the centre of the sun is
where σ is the Stefan's constant.

(A) $\frac{r^2\sigma(t+273)^4}{4\pi R^2}$ (B) $\frac{16\pi^2r^2\sigma t^4}{R^2}$
(C) $\frac{r^2\sigma(t+273)^4}{R^2}$ (D) $\frac{4\pi r^2\sigma t^4}{R^2}$

Open solution

Q71. AIPMT 2007

A black body is at 727°C. It emits energy at a rate which is proportional to

(A) $(1000)^4$ (B) $(1000)^2$ (C) $(727)^4$ (D) $(727)^2$

Open solution

Q72. AIPMT 2006

A black body at 1227°C emits radiations with maximum intensity at a wavelength of 5000 $\mathring{A}$. If the temperature of the body is increased by 1000°C, the maximum intensity will be observed at

(A) 3000 $\mathring{A}$ (B) 4000 $\mathring{A}$
(C) 5000 $\mathring{A}$ (D) 6000 $\mathring{A}$

Open solution

Q73. AIPMT 2005

Which of the following rods, (given radius r and length l) each made of the same material and whose ends are maintained at the same temperature will conduct most heat ?

(A) r = r_0, $l = l_0$ (B) r = 2r_0, $l = l_0$
(C) r = r_0, $l = 2l_0$ (D) r = 2r_0, $l = 2l_0$

Open solution

Q74. AIPMT 2004

If λ_m denotes the wavelength at which the radioactive amission from a black body at a temperature TK is maximum, then

(A) $\lambda_m \propto T^4$ (B) λ_m is independent of T
(C) $\lambda_m \propto$ T (D) $\lambda_m \propto T^{-1}$

Open solution

Q75. AIPMT 2003

Consider a compound slab consisting of two different materials having equal thicknesses and thermal conductivities K and 2K, respectively. The equivalent thermal conductivity of the slab is

(A) $\frac{2}{3}K$ (B) $\sqrt{2}K$ (C) 3 K (D) $\frac{4}{3}K$

Open solution

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Q76. AIPMT 2002

Unit of Stefan's constant is

(A) watt m^2 K^4 (B) watt m^2/K^4 (C) watt/m^2 K (D) watt/m^2K^4

Open solution

Q77. AIPMT 2002

Consider two rods of same length and different specific heats (S_1, S_2), conductivities (K_1, K_2) and area of cross-sections (A_1, A_2) and both having temperatures T_1 and T_2 at their ends. If rate of loss of heat due to conduction is equal, then

(A) $K_1A_1 = K_2A_2$ (B) $\frac{K_1A_1}{S_1} = \frac{K_2A_2}{S_2}$

(C) $K_2A_1 = K_1A_2$ (D) $\frac{K_2A_1}{S_2} = \frac{K_1A_2}{S_1}$

Open solution

Q78. AIPMT 2002

For a black body at temperature 727^oC, its radiating power is 60 watt and temperature of surrounding is 227^oC. If temperature of black body is changed to 1227^oC then its radiating power will be

(A) 304 W (B) 320 W (C) 240 W (D) 120 W

Open solution

Q79. AIPMT 2002

Which of the following is best close to an ideal black body?

(A) black lamp

(B) cavity maintained at constant temperature

(C) platinium black

(D) a lump of charcoal heated to high temperature.

Open solution

Q80. AIPMT 2002

The Wien's displacement law express relation between

(A) wavelength corresponding to maximum energy and temperature

(B) radiation energy and wavelength

(C) temperature and wavelength

(D) colour of light and temperature.

Open solution

Q81. AIPMT 2001

A cylindrical rod having temperature T_1 and T_2 at its end. The rate of flow of heat Q_1 cal/sec. If all the linear dimension are doubled keeping temperature constant, then rate of flow of heat Q_2 will be

(A) $4Q_1$ (B) $2Q_1$ (C) $\frac{Q_1}{4}$ (D) $\frac{Q_1}{2}$

Open solution

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Q82. AIPMT 2000

A black body has maximum wavelength λ_m at 2000 K. Its corresponding wavelength at 3000 K will be

(A) $\frac{3}{2}\lambda_m$ (B) $\frac{2}{3}\lambda_m$ (C) $\frac{16}{81}\lambda_m$ (D) $\frac{81}{16}\lambda_m$

Open solution

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Answer Key

1. **A**	2. **B**	3. **D**	4. **A**	5. **D**
6. **C**	7. **A**	8. **A**	9. **B**	10. **D**
11. **C**	12. **A**	13. **B**	14. **A**	15. **A**
16. **B**	17. **B**	18. **B**	19. **A**	20. **C**
21. **B**	22. **B**	23. **C**	24. **C**	25. **D**
26. **D**	27. **A**	28. **C**	29. **C**	30. **C**
31. **C**	32. **C**	33. **B**	34. **B**	35. **B**
36. **B**	37. **B**	38. **A**	39. **B**	40. **B**
41. **D**	42. **D**	43. **D**	44. **D**	45. **B**
46. **A**	47. **D**	48. **A**	49. **B**	50. **C**
51. **D**	52. **A**	53. **B**	54. **A**	55. **A**
56. **A**	57. **D**	58. **B**	59. **C**	60. **D**
61. **A**	62. **D**	63. **A**	64. **B**	65. **A**
66. **C**	67. **B**	68. **C**	69. **D**	70. **C**
71. **A**	72. **A**	73. **B**	74. **D**	75. **D**
76. **D**	77. **A**	78. **B**	79. **B**	80. **A**
81. **B**	82. **B**			

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